



Exercise 4

Problem 1 : Biphase function

Consider the following shaping filter

$$g(t) = \begin{cases} 1/\sqrt{T_s} & \text{for } t \in [0, T_s/2[\\ -1/\sqrt{T_s} & \text{for } t \in [T_s/2, T_s[\\ 0 & \text{otherwise} \end{cases}$$

Questions :

1. Is $g(\cdot)$ a Nyquist filter?
2. By taking into account only the main lobe, compute the bandwidth of $g(\cdot)$.
3. Is $g(\cdot)$ a square-root Nyquist filter?
4. Deduce a sufficient condition for a filter to be simultaneously Nyquist and square-root Nyquist.

Problem 2 : Nyquist criterion

In this problem, you will prove the Nyquist criterion as presented in the lecture notes (2.29). The proof relies on the following fact concerning periodic functions: Let T be a positive real number and $f: \mathbb{R} \rightarrow \mathbb{R}$ be a periodic function such that $f(t) = f(t + T), \forall t \in \mathbb{R}$. Under certain mild assumptions, $f(\cdot)$ can be expanded into a Fourier series of the form

$$f(t) = \sum_{l \in \mathbb{Z}} c_l e^{2\pi i l t / T} \quad (1)$$

where the coefficients c_l are given by

$$c_l = \frac{1}{T} \int_0^T f(t) e^{-2\pi i l t / T} dt. \quad (2)$$

We begin the exercise by proving the following fact: Let $f(\cdot)$ be given by its Fourier series as in (1). Then $f(\cdot)$ is constant if and only if c_l is zero for all $l \neq 0$. Clearly, this condition is sufficient. Show that the condition is necessary by following the steps below.

1. Find the derivative $f'(\cdot)$ of $f(\cdot)$ with respect to t .
2. Show that $\int_0^T e^{2\pi i k t / T} f'(t) dt = 2\pi i k c_k$.
3. Conclude from the previous question that when $f(\cdot)$ is constant, all coefficients c_l , for $l \neq 0$, must be zero.

To apply the result from step 3, we need to establish the Fourier series of $\sum_{l \in \mathbb{Z}} P(f - l/T_s)$, where $P(\cdot)$ represents the Fourier transform of the impulse response $p(\cdot)$. To this end:

4. Show that $\sum_{l \in \mathbb{Z}} P(f - l/T_s)$ is periodic with period $1/T_s$.
5. Using (2), show that the l -th Fourier coefficient c_l of $\sum_{l \in \mathbb{Z}} P(f - l/T_s)$ is $T_s p(lT_s)$.
Hint: In evaluating (2), interchange the integral and summation, and use the change of variable $f' = f - l/T_s$. This should allow you to recognize the inverse Fourier transform and recover $c_l = T_s p(lT_s)$.

Finally, conclude the Nyquist criterion (2.29) from steps 3 and 5.

Problem 3: from OOK modulation to PAM modulation

Consider a constellation with the two equiprobable symbols

$$s^{(0)} = A \quad \text{and} \quad s^{(1)} = -\theta A$$

where A is a positive real number and θ is a real number between 0 and 1.

At each discrete time n , the receiver observes the symbol

$$z_n = s_n + w_n$$

where

- s_n is a symbol of the constellation defined above.
- w_n is an i.i.d. Gaussian noise of mean zero and variance $N_0/2$.

Questions :

1. Describe the decision regions. What is the threshold separating the decision regions corresponding to $s^{(0)}$ and to $s^{(1)}$?
2. Express the bit error probability as a function of θ and E_b/N_0 .
3. Find the value of $\theta \in [0, 1]$ that minimizes the bit error probability. What is the corresponding constellation?
4. Find the value of $\theta \in [0, 1]$ that maximizes the bit error probability. What is the corresponding constellation? What is the loss in RSB with respect to the constellation found in the previous question?
5. Without calculations, find the bit error probability of the constellation formed by 0 and $2A$. What is the transformation needed to obtain the 2-PAM? Is the bit error probability invariant with respect to this transformation?